

Relativistic Beam Paths as Observation Geometry

From synchrotron phase space to typed detector invariants

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Status. This document supersedes the bilingual working draft v1.2. It is migrated to GO Core v0.2 and the SI-HEP Quantity Passport v0.5. The physical reference is standard Run 3 proton-proton operation; on the document date the LHC is in Long Shutdown 3. No new accelerator model or particle-physics prediction is claimed.

Abstract

An accelerator is not observed as one visible trajectory. Its hidden state contains a lattice-constrained reference orbit, time-dependent beam distributions, RF and betatron phases, magnet settings, collision states, and detector response variables. We type the corresponding chain as

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t,$$

where reduction, ideal observation, stochastic measurement, and estimation are distinct maps. The SI-HEP boundary is explicit: $p_{\text{SI}}^\mu = (E/c, \mathbf{p})$, $\Pi^\mu = (E, c\mathbf{p})$, and the HEP numerical four-vector is $\Pi^\mu/\mathcal{E}_0$. This removes the ambiguity in $E^2 = \mathbf{p}^2 + m^2$. For charged beam transport, SI magnetic rigidity and HEP momentum are connected by the exact declared-unit identity

$$p[\text{GeV}/c] = 0.299792458 z B\rho[\text{T m}].$$

At a Run 3 reference total energy of 6.8 TeV, the proton has $\gamma \approx 7.2474 \times 10^3$, $1 - \beta \approx 9.5194 \times 10^{-9}$, and $B\rho \approx 2.26824 \times 10^4 \text{ T m}$. The manuscript separates Lorentz invariants, longitudinal-boost covariants, symplectic invariants, operational observables, and estimator outputs. In particular, geometric emittance is invariant only under a specified ideal symplectic transport, while normalized emittance is the relevant ideal adiabatic invariant during acceleration. Equal-time Lorentz contraction and Penrose–Terrell camera appearance are retained as different observation maps, neither being a direct imaging model for LHC protons.

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1 Scope and claim boundary

The manuscript is a typed interface between standard accelerator physics and Geometry of Observation. It is not:

- a lattice design, collimation, impedance, beam-loss, or machine-protection model;
- a replacement for tracking, beam-beam, or detector simulation;
- a hard-body model of the proton;
- a detector reconstruction algorithm;
- a claim that a hadron's Lorentz-contracted equal-time profile is photographed in the LHC;
- a new Standard Model result.

The strict contribution is infrastructural: it identifies the objects, maps, unit charts, transformation groups, and claim conditions needed to state accelerator observations without type or dimension gaps.

2 Observation protocol

Definition 2.1 (Hidden accelerator state). A minimal hidden state at machine time t is a record

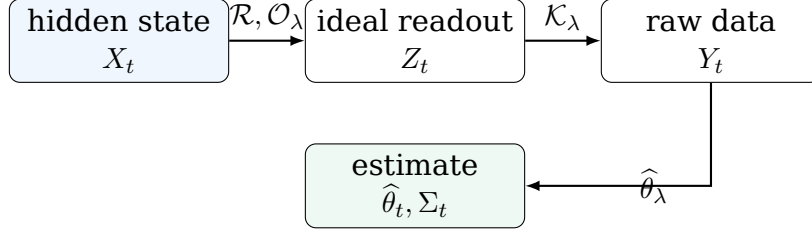
$$X_t = (\Gamma_0, \mathcal{L}_{\text{mag}}(t), \Phi_{\text{RF}}(t), f_1(t), f_2(t), \mathcal{C}_t, \mathcal{D}_t),$$

where Γ_0 is a design orbit, $\mathcal{L}_{\text{mag}}$ is the magnet lattice state, Φ_{RF} is the RF state, f_1, f_2 are beam phase-space distributions, $\mathcal{C}_t$ is a collision-event state when a crossing occurs, and $\mathcal{D}_t$ contains detector-response latent variables.

An observation protocol λ contains a frame, clock, calibration, spatial and temporal resolutions, trigger and selection rules, a deterministic response, a stochastic channel, and an estimator. Its typed chain is

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} Y_t \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

Here $\mathcal{K}_\lambda$ denotes a Markov kernel. It must not be merged with the deterministic map $\mathcal{O}_\lambda$, and $\hat{\theta}_\lambda$ must not be identified with the underlying measurand.



3 Reference machine passport

The numerical audit uses the CERN public Run 3 reference:

$$E_{\text{beam}} = 6.8 \text{ TeV}, \quad \sqrt{s} = 13.6 \text{ TeV}$$

for equal, head-on proton beams. The public circumference $C = 26659 \text{ m}$, roughly 2500 bunches per proton beam, and roughly 1.6×10^{11} protons per bunch are rounded or approximate operational descriptors, not defining constants.

Boundary 3.1 (Epoch). The LHC entered Long Shutdown 3 in June 2026. Therefore 6.8 TeV in this document is a Run 3 standard-operation reference, not a statement that a beam was circulating on 28 July 2026.

Every machine record must include

(facility, species, charge state, run, mode, epoch, status, value, unit, precision, source).

The design energy, an achieved beam energy, a special low-energy run, and an HL-LHC target are different records even when they share a unit.

4 SI-HEP representation boundary

Adopt $\eta = \text{diag}(1, -1, -1, -1)$. The SI momentum four-vector and energy four-vector are

$$p_{\text{SI}}^\mu = (E/c, \mathbf{p}_{\text{SI}}), \quad \Pi^\mu = c p_{\text{SI}}^\mu = (E, c\mathbf{p}_{\text{SI}}).$$

With energy anchor $\mathcal{E}_0 = 1 \text{ GeV}$, define the HEP coordinate

$$P_{\text{HEP}}^\mu = \frac{\Pi^\mu}{\mathcal{E}_0}.$$

The mass shell has three equivalent representations:

$$p_{\text{SI}}^2 = m^2 c^2, \quad \Pi^2 = (mc^2)^2, \quad P_{\text{HEP}}^2 = \left(\frac{mc^2}{\mathcal{E}_0} \right)^2.$$

The shorthand $E^2 = \mathbf{p}^2 + m^2$ is used only in the last chart. In the other charts, omitted factors of c change dimensions. The complete conversion contract is specified in the companion *SI-HEP Quantity Passport v0.5*.

5 Relativistic beam kinematics

Let

$$\mu_p := m_p c^2 = 0.93827208943(29) \text{ GeV}$$

from CODATA 2022, and let E denote proton total energy. Define the energy-valued momentum $\pi := pc$. Then

$$E^2 = \pi^2 + \mu_p^2, \quad \gamma = \frac{E}{\mu_p}, \quad \beta = \frac{\pi}{E}.$$

At high γ , direct subtraction of β from one loses floating-point significance. The stable identity is

$$1 - \beta = \frac{1 - \beta^2}{1 + \beta} = \frac{\gamma^{-2}}{1 + \beta}.$$

This quantity, rather than a rounded decimal for β , controls the speed deficit:

$$c - v = c(1 - \beta).$$

Proposition 5.1 (Equal head-on beams). *For two equal beams with four-momenta $\Pi_1 = (E, \pi)$ and $\Pi_2 = (E, -\pi)$,*

$$s = (\Pi_1 + \Pi_2)^2 = 4E^2, \quad \sqrt{s} = 2E.$$

Proof. The total spatial momentum vanishes, while the total energy is $2E$. $\square$

The result assumes equal, exactly head-on beams and neglects event-wise energy spread and crossing-angle corrections. Those effects belong to the event and beam-distribution passports.

6 Frame slices and light-cone observations

For a boost of rapidity Y along the beam axis,

$$\begin{aligned} ct' &= \gamma(ct - \beta z), & z' &= \gamma(z - \beta ct), \\ x' &= x, & y' &= y. \end{aligned}$$

The interval

$$\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$$

is Lorentz invariant. A rest-defined longitudinal proper length L_0 , evaluated between endpoints simultaneous in the laboratory, has laboratory equal-time length $L_{\parallel} = L_0/\gamma$. This is a relation between simultaneity slices, not a material compression law for a quantum proton.

In light-front coordinates

$$x^{\pm} = \frac{ct \pm z}{\sqrt{2}},$$

the same boost acts as

$$x^{+'} = e^{-Y} x^+, \quad x^{-'} = e^Y x^-.$$

A camera image is a different map. A source worldline $X^{\mu}(\xi, \tau)$ contributes to an image at observer event O^{μ} only if

$$(O - X(\xi, \tau_e))^2 = 0, \quad \tau_e < t_{\text{obs}}.$$

The incoming null direction is then projected to an image plane. Thus:

$$\text{equal-time boost} \neq \text{past-light-cone image}.$$

The Penrose–Terrell effect concerns the second map. It is not a detector-observable model for individual LHC protons.

7 Magnetic rigidity and path geometry

In SI,

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \frac{dE}{dt} = q \mathbf{E} \cdot \mathbf{v}.$$

A transverse magnetic field bends the orbit but does no work. For local curvature radius ρ ,

$$|\mathbf{p}|_{\text{SI}} = |q| B \rho.$$

With $q = ze$, the exact declared-unit bridge is

$$\boxed{p[\text{GeV}/c] = 0.299792458 z B \rho[\text{T m}].}$$

For a Run 3 reference proton,

$$p = 6799.999935268 \text{ GeV}/c, \quad B \rho = 22682.358258 \text{ T m}.$$

Boundary 7.1 (Local versus global geometry). The circumference contains straight insertions and does not determine the dipole bending radius by $C/(2\pi)$. The local field-to-curvature map requires the lattice geometry. Conversely, the revolution frequency may use the total closed-orbit length.

8 RF phase and longitudinal state

For an idealized RF gap,

$$\Delta E = q V_{\text{eff}} \sin \phi_s.$$

For $q = ze$,

$$\Delta E[\text{eV}] = z V_{\text{eff}}[\text{V}] \sin \phi_s.$$

The unit conversion is exact; the effective-voltage and thin-gap model is not.

A common dimensionless longitudinal coordinate is

$$\delta = \frac{p - p_0}{p_0}.$$

The phase ϕ and δ are not direct positions and momenta until a canonical normalization is declared. Consequently, a plotted RF “bucket area” is not comparable across conventions without the Jacobian and reference energy. The observation protocol must record the RF harmonic, synchronous phase, voltage convention, longitudinal coordinate convention, and clock.

9 Transverse optics and emittance

In one transverse plane, linearized motion about the design orbit is

$$x''(s) + K(s)x(s) = 0, \quad [K] = L^{-2}.$$

Here x has dimension L , s has dimension L , and $x' = dx/ds$ is dimensionless. A first-order transport map $\mathbf{u}_2 = M\mathbf{u}_1$, $\mathbf{u} = (x, x')^\top$, is symplectic when

$$M^\top J M = J, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For a centered beam with covariance matrix Σ , define the geometric RMS emittance

$$\varepsilon := \sqrt{\det \Sigma}, \quad \Sigma = \varepsilon \begin{pmatrix} \beta_T & -\alpha_T \\ -\alpha_T & \gamma_T \end{pmatrix}, \quad \beta_T \gamma_T - \alpha_T^2 = 1.$$

The dimensions are

$$[\beta_T] = L, \quad [\alpha_T] = 1, \quad [\gamma_T] = L^{-1}, \quad [\varepsilon] = L.$$

Proposition 9.1 (Qualified geometric-emittance invariance). *If $\Sigma_2 = M\Sigma_1 M^\top$ and M is a 2×2 symplectic map in the same canonical coordinate convention, then $\det \Sigma_2 = \det \Sigma_1$ and $\varepsilon_2 = \varepsilon_1$.*

Proof. A 2×2 symplectic matrix has $\det M = 1$. Hence $\det \Sigma_2 = (\det M)^2 \det \Sigma_1 = \det \Sigma_1$. $\square$

During ideal acceleration the geometric emittance in trace-space coordinates undergoes adiabatic damping. The corresponding normalized emittance is

$$\varepsilon_n = \beta_{\text{rel}} \gamma_{\text{rel}} \varepsilon.$$

Its invariance still requires ideal uncoupled transport and excludes scattering, radiation, noise, nonlinear filamentation, collective effects, and losses. The word “invariant” without this hypothesis list is rejected.

10 Bunches as hidden distributions

Let $u = (x, p_x, y, p_y, z, \delta)$ denote a declared six-dimensional coordinate chart and let

$$f_t(u) \geq 0, \quad \int f_t(u) d\mu_\Gamma(u) = 1$$

be a normalized hidden beam law with a typed phase-space measure. An intensity density uses a different normalization, $\int f_t d\mu_\Gamma = N_b$.

Profile monitors observe marginals or convolutions, for example

$$\rho_x^{\text{ideal}}(x) = \int f_t(u) d\mu_{\Gamma \setminus x}, \quad Y_x \sim \mathcal{K}_\lambda(\rho_x^{\text{ideal}}).$$

Equal one-dimensional marginals do not identify the joint law. Different transverse correlations, tails, halo populations, and loss risks can therefore agree at the profile level.

11 Collision and detector reconstruction

For reconstructed final-state energy four-vectors $\Pi_i^\mu = (E_i, \boldsymbol{\pi}_i)$,

$$\mu_{\text{inv}}^2 = \left(\sum_i E_i \right)^2 - \left\| \sum_i \boldsymbol{\pi}_i \right\|^2.$$

The same physical invariant in SI momentum coordinates is

$$m_{\text{inv}}^2 c^4 = \left(\sum_i E_i \right)^2 - c^2 \left\| \sum_i \mathbf{p}_i \right\|^2.$$

Detector data do not provide the exact Π_i^μ . Tracks, clusters, timing, calibration constants, object identification, and selections enter an estimator

$$\hat{\mu}_{\text{inv}} = \hat{\theta}_\lambda(Y), \quad \Sigma_{\hat{\mu}} \text{ or a likelihood interval.}$$

For polar angle θ ,

$$p_T = \|\mathbf{p}_\perp\|, \quad \eta = -\ln \tan \frac{\theta}{2}.$$

The logarithm is dimensionless. Under a longitudinal boost, p_T is unchanged. True rapidity

$$y = \frac{1}{2} \ln \frac{E + \pi_z}{E - \pi_z}$$

shifts additively, so rapidity differences are boost-invariant. Pseudorapidity equals rapidity only in the massless limit; it is not an unrestricted Lorentz scalar.

12 Transformation classes

Class	Object	Correct status
Lorentz	Π^2, μ_{inv}	invariant under proper Lorentz transformations
Longitudinal boost	$p_T, y, \Delta y$	p_T unchanged, y shifts, Δy invariant
Symplectic transport	$\varepsilon = \sqrt{\det \Sigma}$	invariant under the stated ideal map and coordinate convention
Adiabatic acceleration	ε_n	ideal normalized-emittance invariant under additional hypotheses
Unit chart	physical Q	same quantity; numerical coordinate transforms covariantly
Detector reconstruction	$\hat{\mu}_{\text{inv}}$	estimator with calibration and uncertainty, not an exact invariant
Magnet setting	$B\rho$	frame- and species-dependent rigidity relation, not Lorentz invariant

13 Non-identifiability

For a deterministic observation $y = \mathcal{O}_\lambda(x)$, the exact fiber is

$$\mathcal{F}_y = \{x : \mathcal{O}_\lambda(x) = y\}.$$

For a stochastic channel with likelihood $k_\lambda(y | x)$, the compatibility set becomes

$$\mathcal{F}_{y,\lambda} = \{x : k_\lambda(y | x) > 0\},$$

or, more usefully, a likelihood or posterior level set. A “size of the fiber” is undefined until a metric, measure, prior, threshold, and quotient by representation symmetries are specified.

This distinction blocks three invalid inferences:

- (i) a beam profile does not determine the full phase-space law;
- (ii) a reconstructed invariant mass does not determine a unique partonic history;
- (iii) an event display is not the collision event itself.

14 Reproducible Run 3 numerical audit

The following values use the exact SI definitions of c, h, e , the CODATA 2022 proton mass energy, the CERN public 6.8 TeV reference, and the rounded public circumference.

Quantity	Value	Status
Reference proton total energy	6800 GeV	operational reference
Proton mass energy	0.93827208943 GeV	CODATA 2022
Lorentz factor γ	$7.24736467876 \times 10^3$	derived
Speed ratio β	0.99999999048060	derived
Stable $1 - \beta$	$9.519404390069 \times 10^{-9}$	derived
Speed deficit $c - v$	$2.853845641 \text{ m s}^{-1}$	derived
Momentum	6799.999935268 GeV/ c	derived
Magnetic rigidity $B\rho$	22682.358258 T m	derived
Revolution frequency	11245.450135 Hz	rounded circumference
Symmetric head-on $\sqrt{s}$	13600 GeV	derived

The revolution-frequency value has more displayed digits only as a regression output; its physical precision is limited by the rounded circumference and the orbit definition. The approximate public bunch count and population would imply a beam-energy scale of about $4.36 \times 10^8 \text{ J}$, but this is not an event-by-event or fill-specific measurement.

15 Claim audit

Statement	Status	Required hypotheses or reason
Mass-shell equivalence across SI/HEP charts	theorem	explicit c -map and common signature
Run 3 numerical audit	derived check	declared source values and epoch
Geometric-emittance invariance	proposition	linear symplectic map, fixed canonical convention
Normalized-emittance invariance	ideal model	adiabatic acceleration and excluded nonideal effects
Equal-time longitudinal contraction	standard kinematics	inertial frames and declared simultaneity slices
Penrose–Terrell appearance	standard optical result	past-light-cone image of an extended source
Direct optical image of an LHC proton	rejected	no such detector channel is modelled
New LHC or particle-physics prediction	not claimed	reference/infrastructure module only

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